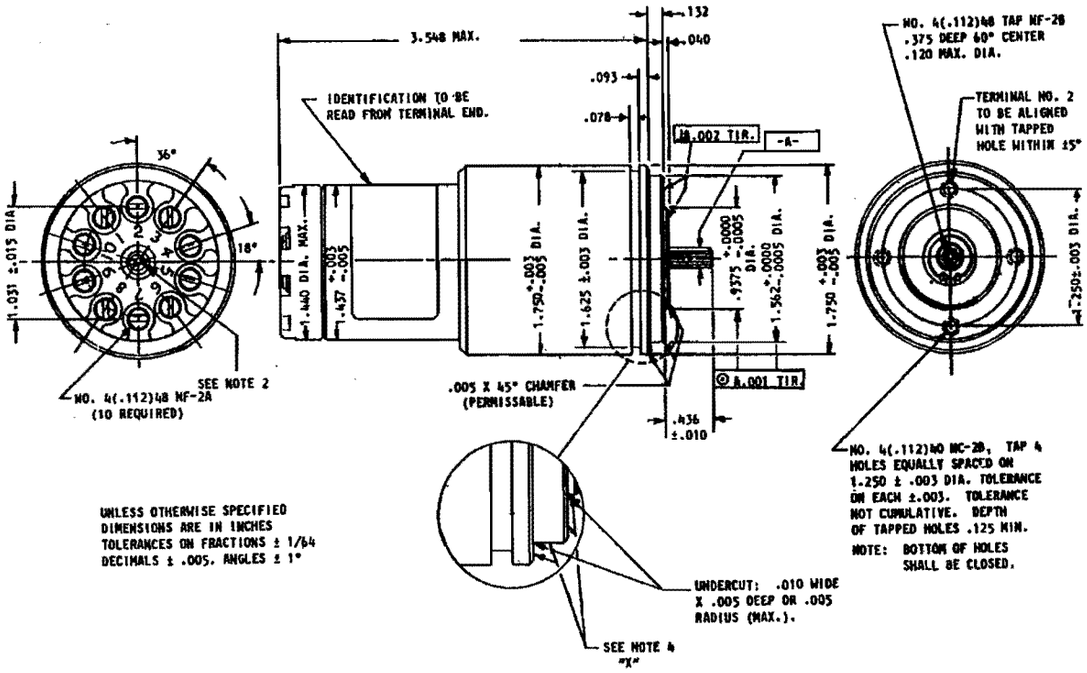
MIL-S-22820/7B
1 November 1972
SUPERSEDING
MIL-S-22820/7A
3 March 1966

MILITARY SPECIFICATION SHEET
SERVOMOTOR TACHOMETER-GENERATOR, TYPE 18SM-RG4d

This Specification is Mandatory for use by all Departments and Agencies of the Department of Defense.
THE COMPLETE REQUIREMENTS FOR PROCURING THE SERVOMOTOR-TACHOMETER GENERATOR DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF MIL-S-22820.



UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES ON FRACTIONS ± 1/64
DECIMALS ± .005. ANGLES ± 1°

SPUR GEAR DATA	
TEETH	15
PITCH	96
PITCH DIAMETER	.1562 +.0000 -.0005
OUTSIDE DIAMETER	.1770 +.0000 -.0002
PRESSURE ANGLE	20°
TOOTH FORM	FULL DEPTH INVOLUTE
DESIGN TO MEET AGMA 207.04 (ASA-B-6,7-1956) '20 DEGREE INVOLUTE FINE PITCH SYSTEM' PRECISION GRADE 1[C] TESTED TO AGMA 236.04 (ASA-B-6-11-1956) INSPECTION OF FINE GEARS.	

NOTES:
1. PERPENDICULARITY AND CONCENTRICITY SHALL BE MEASURED WITH THE UNIT IN A VERTICAL POSITION SUPPORTED BY THE SHAFT, THEN THE HOUSING SHALL BE ROTATED.
2. RUNOUT OF FEATURE "A" SHALL NOT EXCEED .0014 TIR.
3. END PLAY AND RADIAL PLAY SHALL BE MEASURED DURING THE REVERSAL OF THE FOLLOWING LOADS:
   END PLAY - 1 POUND RADIAL PLAY - 1 POUND
4. THE DESIGN OF THIS UNIT SHALL PROVIDE FOR TERMINAL BLOCK TO REMAIN FIXED WHEN TERMINAL SCREWS ARE REMOVED.
5. THE END CAP DESIGN SHALL BE SUCH THAT THE TERMINAL LUG MS-17182-1 MAY BE INSTALLED WITH THE BARREL INBOARD OR 180° POSITIVE OUTBOARD. RECESSES FOR LUGS SHALL BE OF WIDTH AND DEPTH TO PREVENT TURNING OF THE LUGS.
6. FINISH OF HOUSING SHALL BE FLAT BLACK AS PER MIL-C-18924, EXCEPT FOR SURFACES MARKED "X" WHICH SHALL BE BRIGHT METALLIC FINISH.
7. DIMENSIONS AND TOLERANCES COMPLY WITH USASI Y14.5.

FIGURE 1

FSC 6105

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TABLE I
SERVOMOTOR-TACHOMETER GENERATOR, TYPE 18SM-RG4d
(Formerly Identified as BUORD Mk 16 Mod 3)

Performance Requirements and Characteristics

Requirements	Value	Unit	Tolerance
Electrical:			
Frequency:			
Rated	400	cps	
Test	400	cps	±1%
Range	380 to 420	cps	
Voltage:			
Reference Phase, Rated	115	volts	
Reference Phase, Test	115	volts	±1%
Reference Phase, Range	115	volts	±10%
Control Phase, Rated (Series)	36	volts	
Control Phase, Test (Series)	36	volts	±1%
Control Phase, Range (Series)	0 to 39.6	volts	
Control Phase, Rated (Parallel)	18	volts	
Control Phase, Range (Parallel)	0 to 19.8	volts	
Current:			
Reference Phase	.200	amperes	maximum
Control Phase (Series)	.634	amperes	maximum
Control Phase (Parallel)	1.17	amperes	maximum
Power:			
Reference Phase	9.1	watts	maximum
Control Phase (Series)	9.1	watts	maximum
Control Phase (Parallel)	8.5	watts	nominal
Synchronous Speed	6000	rpm	
Impedance			
Reference Phase	268+j580	ohms	nominal
Control Phase (Series)	26+j57	ohms	nominal
Power Factor:			
Reference Phase	.42		nominal
Control Phase (Series)	.42		nominal
Effective Resistance:			
Control Phase (Series)	125	ohms	nominal
Control Phase (Parallel)	36	ohms	nominal
Minimum Starting Voltage	1	volt	maximum
Coupling in Control Winding	.7		minimum
Mechanical:			
No-load Speed	4500	rpm	minimum
Stall Torque at 23°C	2.20	ounce-inches	minimum
Rotor Moment of Inertia	5.73	$gm\text{-}cm^2$	maximum
Radial Play	.0008	inch	maximum
End Play	.0035	inch	maximum
	.0010	inch	minimum

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TABLE I (Continued)

Requirements	Value	Unit	Tolerance
Temperature Rise	85	degrees C	maximum
Noise, Mechanical	not excessive		
Time Constant	.017	seconds	nominal
Reversing Time	.029	seconds	nominal
Theoretical Acceleration at Stall	29000	radians/second^2	nominal
Maximum Power Output	1.9	watts	nominal
Weight	21	ounces	maximum
Barrier Coating	See MIL-STD-1334(AS)		

Performance Requirements and Characteristics, Tachometer Generator Section

Requirements	Value	Unit	Tolerance
Electrical:			
Frequency:			
Rated	400	cps	
Test	400	cps	±1%
Range	380 to 420	cps	
Voltage:			
Energizing Winding, Rated	115	volts	
Energizing Winding, Test	115	volts	+1%
Energizing Winding, Range	115	volts	±10%
Current:			
Energizing Winding	.078	amperes	maximum
Energizing Winding	.062	amperes	minimum
Power:			
Energizing Winding	5.6	watts	maximum
Power Factor:			
Energizing Winding	.63		nominal
Effective Resistance:			
Energizing Winding	2645	ohms	nominal
Impedance:			
Energizing Winding	1035+j1276	ohms	nominal
Output Winding	3000	ohms	maximum
Output Winding	2500	ohms	minimum
In-phase Speed Sensitive Transformation Ratio (at 3000 rpm) (See Note 2)	.08056		±2.5%
Phase Shift (at 3000 rpm) (See Note 2)	4.5 lag	degrees	±1.5
In-Phase Linearity (See Note 2)	±0.8	percent	
In-Phase Axis Error (See Note 2)	∓.008	volts	maximum
Zero Speed Output Voltages (See Note 2)			
Total	.013	volts	maximum
Fundamental	.010	volts	maximum
Output Gradient	3.1	volts/1000 rpm	nominal

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Notes:

1. A stall torque requirement during the high ambient temperature (100°C) test of MIL-S-22820 is not specified. Refer also to paragraph 3.32 of the same document.

2. Measured open circuited for values specified in the Table.

3. The servomotor-tachometer generator identified and specified by this specification shall conform to all requirements, and shall be subjected to all tests delineated in MIL-S-22820 with the following exceptions:

a. Salt Atmosphere Resistance - paragraph 3.35
b. Audible Noise, Structureborne - paragraph 3.36
c. Voltage Sensitivity (Tachometer section) - paragraph 3.24
d. Frequency Sensitivity (Tachometer section) - paragraph 3.25
e. Temperature Sensitivity (Tachometer section) - paragraph 3.26
f. Warm-up Time - paragraph 3.29
g. Position Error Voltage - paragraph 3.20

4. In addition to the performance requirements listed in Table I, the type 18SM-RG4d servomotor-tachometer generator shall possess the nominal characteristics listed therein.

5. Type 18SM-RG4d servomotor-tachometer generators shall be capable of operation in ambient temperatures ranging from -55°C to 100°C.

6. The speed range for the 18SM-RG4d is zero to 3600 rpm. The calibration speed for determination of linearity requirements of Table I shall be 3000 rpm. Phase shift and transformation ratio tests shall be conducted at calibration speed.

7. If, during measurement, the ambient temperature is in the range of temperatures defined by 23° ± 5°C, the corrected transformation ratio shall be:

$$TR_c = \frac{TR_o}{1 - \frac{(t_o - 23)}{400}}$$

Where: TR_c = Corrected transformation ratio
TR_o = Measured transformation ratio
t_o = Ambient temperature at which TR_o was measured.

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The corrected transformation ratio shall be within the limits specified in Table I herein. Similarly, the corrected lagging phase shift angle shall be:

$$\phi_c \text{ degrees} = \phi_o \text{ degrees} - 1/8\ (t_o - 23) \text{ degrees}$$

Where: ϕ_c degrees = Corrected lagging phase shift angle
ϕ_o degrees = Measured lagging phase shift angle
t_o = Ambient temperature at which ϕ was measured

The corrected lagging phase shift shall be within the limits specified in Table I herein.

8. Connection Diagram:

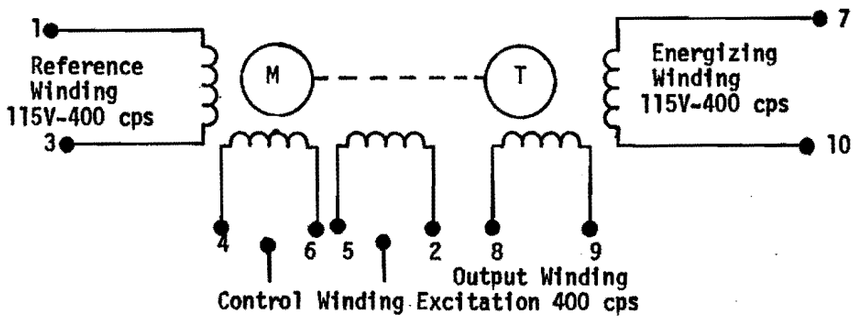


FIGURE 2

a. For CCW rotation the voltage at terminal 2 leads the voltage at terminal 1 by 90 electrical degrees.

b. For series operation of control winding, terminals 5 and 6 are strapped and 36 volts rated is applied across terminals 2 and 4.

c. For parallel operation of control winding, terminals 6 and 2 are strapped, and 18 volts rated is applied across terminals 2 and 4.

d. When shaft is rotated CCW and terminals 9 and 10 are common, the output voltage at terminal 8 is out of phase with the voltage at terminal 7.

9. Sampling, inspection, and preparation for delivery shall be in accordance with MIL-S-22820.

Custodians:
Army - MU
Navy - AS
Air Force - 11
Review Activities:
Army - EL, MU, MI
Navy - SH, AS
Air Force - 11, 17,
User Activities:
Army -
Navy - MC, SH

Preparing Activity:
Navy - AS

Project No. 6105-0103

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